

ZS-F38

The Register Clock Identity: Unique-Ergodicity Selection of the Democratic Register Measure, the Q-Unit Inclusion Chain, and the Conditional Depth-Selection Theorem $t^* = Q$

Registering the ZS-F37 Appendix C.5 Gate as the Register Clock Identity — One Named Identity Behind the Three Open Programmes of the ZS-F33–F36 Line, with the Anti-Numerology Statistical Hierarchy Fixed of Record and Executed in ZS-A32

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Theme / Code: Foundations / ZS-F38 · Registers the ZS-F37 Appendix C.5 gate as (H-CLK) · Companion to ZS-M46 / ZS-M47, ZS-M44, ZS-F36, ZS-A31 · Upstream of ZS-A32 (v1.0, executed) and ZS-M48 / ZS-F39

Version: 1.1 (review-integrated; supersedes v1.0)

Verification: 32/32 exact/numerical checks **PASS** + 3/3 guards (zs_f38_verify_v1_1.py; SymPy exact linear algebra, rational Markov matrices, and polynomial ideals + mpmath 60-digit + NumPy Perron–Frobenius powers) | **Zero fitted parameters** | Theorem F38.T1' (Primitive Clock Ergodicity) mathematics **IMPORTED-PROVEN** + **PROVEN** (Perron–Frobenius; Birkhoff–von Neumann), physical instance **DERIVED-CONDITIONAL** on (H-CLK) $\wedge$ (H-mix); Corollary F38.T1 the deterministic (H-cycle) case; Theorem F38.T2 **IMPORTED-PROVEN** skeleton (Borchers; Wiesbrock; Jones–Longo; Koot) + **DERIVED** bookkeeping with the standard-subspace bridge Lemma F38.L2; Theorem F38.T3 **DERIVED-CONDITIONAL** on the registry {(H-CLK), (H-cycle), KH1–KH4, (H-eval)}, (H-eval) decomposed into three sub-gates (H-eval-N/F/B) whose F-component carries the ZS-A32 “parent injectivity + NDC3” wording (registries reconciled of record) | **the B3 hierarchy debt is NOT closed** (two-gate debt: (H-CLK) discharge, C_UV 1PI) | anti-numerology **statistical hierarchy fixed of record** (PRIMARY $p_{\text{single}} = 0.50\%$; SECONDARY formula-count $2/88 = 2.27\%$; look-elsewhere 28.8%) and **EXECUTED in ZS-A32 v1.0** — verdict-neutral under both decision candidates | (**A**, **Q**, $\dim \mathbf{Z}$) = (35/437, 11, 2) **LOCKED**.

§0. Abstract

ZS-F36 v2.1 terminated the charge-unit line at $\rho_{\Lambda, Z} = (1260/4807) C_{UV} M_K^4$ with three open research programmes: (i) the full parent 1PI factor C_{UV} , (ii) the action-level selection of the register measure $\rho_Q = I_Q/Q$ (ZS-M44: asserted, not derived), and (iii) the metric-scale hierarchy $M_K/\bar{M}_P$ (ZS-A22 barrier B3), whose single pre-registered mechanism candidate is the modular depth $M_K/\bar{M}_P = e^{(-2\pi Q)}$ with the 2π Borchers–Wiesbrock-forced (ZS-A31 §6.3) and $t = Q$ carried as a *NON-CLAIM* through ZS-M46/M47/F37. This paper observes and formalizes that (ii), (iii), the ZS-M46 exact-clock gate CRT-4a, the ZS-F37 C_{int} protocol, and the ZS-F37 Appendix C.5 pre-registered seam-gauge gate are projections of one named identity, which we define and register as the Register Clock Identity (H-CLK): the physical Z-Spin seam transport clock — the ZS-F31 GKLS register clock advancing one slot per unit — is the modular clock of the ZS-M46 seam inclusion (the F37 C.5 holonomy gate $Hol_{\{A_Z\}}(\gamma) = g^{-1} Hol_{\{A_K\}}(\gamma) g$ on Σ_2 , conjoined with CRT-4a). (H-CLK) is condition reuse: it introduces no new registry

entry beyond a bundling of pre-existing gates, and it remains OPEN as a physical identification — this paper registers and consolidates the F37 C.5 gate; it does not* discharge it.

Under (H-CLK) and a mixing condition we prove three results, upgraded in this version per external review. **Theorem F38.T1' (Primitive Clock Ergodicity).** If the one-tick register transition P is an irreducible doubly stochastic (unital) matrix on the $Q = 11$ slots — condition (H-mix) — then its unique stationary measure is $\rho_Q = I_Q/Q$ and the Cesàro time average of every slot observable converges to the democratic average (with direct convergence when P is primitive); the deterministic 11-cycle of v1.0's (H-cycle) is the extreme irreducible-periodic case (**Corollary F38.T1**), and the sector-preserving alternative is the reducible chain whose stationary simplex is exactly the two-parameter sector-weighted family that ZS-M44's arrowhead solve excluded to 0.45% — the corpus's democratic outcome is thus *primitive-mixing evidence*, broader than permutation evidence. Consequently, for every normalized slot vector $|s\rangle$ (equivalently slot projector P_s), $\text{Tr}(\rho_Q P_s) = 1/Q$, and the ZS-M44 slot-normalized register-trace coupling is $\kappa^2 = A/Q$ with ρ_Q obtained as the clock average, not asserted — DERIVED-CONDITIONAL on (H-CLK) $\wedge$ (H-mix). **Theorem F38.T2 (Q-Unit Inclusion Chain).** Under KH1–KH4 the translated algebras $M_k = U(k)MU(-k)$, $k = 0, \dots, Q$, form a nested chain of half-sided modular inclusions whose even subchain is the ZS-M47 Jones–Longo tunnel ($\Gamma = U(2)$); the new **Lemma F38.L2** supplies the standard-subspace bridge — Koot's spectral-subspace criterion is applied to the standard real subspaces $K_k = \text{closure}((M_k)_{sa} \Omega)$ and transported to the algebra chain by the second-quantization functor available in the ZS-M46 Fock realization under KH1–KH4 — and the order is realized exactly in the Hardy model by inner-function divisibility. **Theorem F38.T3 (Conditional Depth Selection).** Under $\{(H-CLK), (H-cycle), KH1-KH4, (H-eval)\}$, one register-clock period is Q modular units; the Borchers commutation dilates the unit seam step on the stage-oriented branch to $e^{(+2\pi Q)}$ at period return, so $\ell_K = \ell_P e^{(2\pi Q)}$, equivalently $*M_K/\bar{M}_P = e^{(-2\pi Q)}$, i.e. $t = Q$. The evaluation condition is decomposed into three falsifiable sub-gates — (H-eval-N) normalization (the unit seam step at $t = 0$ is the reduced Planck length), (H-eval-F) functoriality (modular dilation $\mapsto$ length scaling; carried by the ZS-M47 NDC3 modular-Dirac structure, inheriting parent injectivity via CRT-1), (H-eval-B) branch choice (stage $\mapsto$ length/time, actor $\mapsto$ energy/mass) — and the ZS-A32 v1.0 registry wording “ $\{(H-CLK), (H-cycle), KH1-KH4, \text{parent injectivity, NDC3}\}$ ” is identified of record as the (H-eval-F) inheritance chain spelled out: the two registries are the same. The (H-eval-N/F) discharge target is stated precisely as the modular length functor E_{len} (Appendix D), scoped to ZS-M48/ZS-F39. Finally, Appendix C fixes the anti-numerology statistical hierarchy of record* — PRIMARY $p_{single} = 2 \cdot \text{band} / \ln(\bar{M}_P/H_0) = 0.50\%$ (the target was registered in ZS-A31 v1.2 before the universe existed; the 88-formula universe is the look-elsewhere audit, coverage 28.8%), SECONDARY formula-count $2/88 = 2.27\%$ (the v1.0 literal reading; the two in-band members are the degenerate pair $2\pi \cdot 11 = \pi \cdot 22$, disclosed), sharper 0.079% reported — and records its execution in ZS-A32 v1.0 ($p_{single} = 0.500\%$ analytic, 0.499% at 10^6 draws). The correction is verdict-neutral: both decision candidates pass at $\leq 5\%$. Honest terminus: B3 is not closed — a two-gate debt $\{(H-CLK) \text{ discharge, } C_{UV} \text{ IPI}\}$ remains; this paper's contribution is the conversion of a loose depth numerology into a named conditional theorem, a falsification registry, and an executed pre-registered statistical protocol. Verification 32/32 PASS + 3/3 guards; zero fitted parameters; $(A, Q, \dim Z^*) = (35/437, 11, 2)$ LOCKED.

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact machine verification; no undischarged assumption.
IMPORTED-PROVEN	Proven in the external literature and used without re-proof; cited.
DERIVED	Follows from PROVEN / IMPORTED-PROVEN results by stated steps; no new parameter.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions ((H-CLK), (H-cycle), KH1–KH4, (H-eval)).
DERIVED-BY-INHERITANCE	Uses an upstream corpus result, not re-proven here; inherits its stated caveats.
CONFIRMED-NUMERICALLY	Established by exact computation to a stated, input-limited precision (support, not proof).
HYPOTHESIS-strong / -weak	Structurally motivated / pre-registered numerical relation; mechanism not derived.
CONSISTENCY-CHECK / GUARD	A numerical or protocol check that does NOT by itself certify a theorem.
PRE-REGISTERED	A statistical protocol fixed in advance of execution (execution in a named successor).
NON-CLAIM	Explicitly outside this paper's scope (deferred to a named successor).
OPEN	A genuine, well-posed gap.

§1. Introduction

Three research programmes were left open by the terminal ZS-F36 v2.1: the full parent 1PI factor C_{UV} , the action-derived register measure, and the metric-scale hierarchy $M_K/\bar{M}_P$ (the ZS-A22 barrier B3). Each was declared "a separate research programme." The present paper begins from a structural observation: the second and third programmes, together with the ZS-M46 exact-clock gate CRT-4a, the ZS-F37 finite C_{int} protocol, and the seam-gauge gate that ZS-F37 Appendix C.5 pre-registered under the name ZS-F38, are not five independent problems. Each asks, in its own coordinates, whether **one clock ticks**: whether the Z-Spin seam transport step — proven by ZS-M46 (Theorem M46.3A) to be the unit translation $u \circ f = u + 1$ on the canonical Abel cover — is the same tick as the register's slot advance.

This paper names that identification, proves what follows from it by elementary and imported-proven mathematics, and states honestly what does not follow. The mathematics is deliberately light: finite unital clock ergodicity (Theorem F38.T1', with the deterministic cycle case as Corollary F38.T1), the Borchers–Wiesbrock translation chain with Koot's 2025 relative-position criterion (Theorem F38.T2), and the Borchers commutation relation applied at one register period (Theorem F38.T3). The physics is deliberately conditional: every theorem's first sentence carries its named conditions, following the ZS-F37 discipline. The paper's net effect on the corpus ledger is a **compression**: three OPEN programmes and two OPEN gates become one bundled identification (H-CLK), one structural condition (H-cycle), one evaluation condition (H-eval), and one computation (C_{UV}) — with the statistical protocol for the depth candidate fixed in advance (Appendix C) so that ZS-A32 can execute it without post-hoc freedom.

v1.1 scope note. This version integrates a detailed external review of v1.0 together with the now-existing ZS-A32 v1.0 (which executed the v1.0 Appendix C design). Five review corrections are implemented: the statistical-hierarchy fix of Appendix C (verdict-neutral; §10, Appendix C), the (H-eval) decomposition reconciling the ZS-A32 registry wording (§6), the slot-projector narrowing of the T1 coupling statement (§4), the standard-subspace bridge for Koot's criterion (§5, Lemma F38.L2), and the retitling of the F37

C.5 relation from “executing” to “registering” (title; §1). Two review strengthenings are implemented: Theorem F38.T1’ generalizing the measure selection from permutations to irreducible unital Markov transitions (§4), and the E_len functor proof target (Appendix D). No theorem conclusion, numerical value, or epistemic verdict changes.

Per the ZS-A31 v1.2 discipline, the Planck convention is fixed to the reduced $\bar{M}_P$ before any comparison; the observed 5.4% gap between $M_{eff}/\bar{M}_P$ and $e^{(-2\pi Q)}$ must be accounted for only by a computed C_{UV} , never absorbed into Q ; and the phenomenological reading $M_{eff}/\bar{M}_P = e^{(-2\pi Q)}$ is explicitly *not* the candidate — the candidate is the emergent-metric reading $M_K/\bar{M}_P = e^{(-2\pi Q)}$ with $M_{eff} = C_{UV}^{(1/4)} M_K$.

§2. Locked inputs and the two clocks

Locked inputs (ZS-M1; ZS-F31 App. D; ZS-M6; ZS-M44 v1.6): $A = 35/437$, $Q = 11$, ($\dim Z$, $\dim X$, $\dim Y$) = (2, 3, 6), $\kappa^2 = A/Q = 35/4807$; $z = 0.4382829367 + 0.3605924719i$, $\lambda = (i\pi/2)z$, $\mu = -\ln|\lambda| = 0.1148346250$, $\theta = \arg \lambda^* = 2.2592495540$. All are re-verified to locked digits in the suite (KO1–KO2).

Clock one (the seam). ZS-M46 (Theorem M46.3A, PROVEN) shows that on the canonical Abel cover $u = \text{Log } \chi / \text{Log } \lambda$ the *Z-Spin iteration is the unit translation* $u \circ f = u + 1$; the modular dilation $\Delta^{(it)}$ of the completed standard pair is its positive-energy completion (Borchers), and ZS-M47 (Theorem M47 (I), DERIVED) identifies the canonical endomorphism as the doubled seam step $\Gamma = J_N J_M = U(2)$, the Jones–Longo tunnel being the depth chain. The suite re-verifies the Abel increment $u(fz) - u(z) = 1$ at twelve independent orbit points to $< 10^{-12}$ (KO3), by the branch-free ratio identity $\chi(fz) = \lambda\chi(z)$.

Clock two (the register). ZS-F31 constructs the exact modular GKLS spectrum of the register dynamics — the register clock — whose tick advances the Q -slot register by one slot per unit by construction. ZS-M46 CRT-4a (OPEN) asks whether the transported seam clock equals the core register clock up to a central $(\mathbb{T} \cdot 1)$ cocycle; ZS-F37 Appendix C.5 pre-registered, under the code ZS-F38, the gauge form of the same question: $A_Z|_{\Sigma_2}$ gauge-equivalent to $A_K|_{\Sigma_2}$, i.e. $\text{Hol}_\gamma\{A_Z\}(\gamma) = g^{-1} \text{Hol}_\gamma\{A_K\}(\gamma) g$ for physical seam loops $\gamma \subset \Sigma_2$. A central cocycle does not rescale a flow, so CRT-4a, if it holds, fixes the two clocks to the **same speed**: one slot per unit of seam modular time.

§3. The Register Clock Identity (H-CLK) and the cycle condition (H-cycle)

Definition F38.D1 (Register Clock Identity, (H-CLK)) [OPEN as a physical identification; condition reuse]. (H-CLK) is the conjunction of (a) the ZS-F37 C.5 seam-gauge gate $(A_Z|_{\Sigma_2} \cong A_K|_{\Sigma_2}$ as above) and (b) ZS-M46 CRT-4a (central-cocycle clock equality), read with the ZS-F31 GKLS normalization (one slot advance per unit register time). No component is new: (H-CLK) bundles pre-existing OPEN gates and adds no registry entry. Its content in one sentence: **the register's tick is the seam's unit modular time.**

Definition F38.D2 (Mixing and cycle conditions) [OPEN; partially supported]. (H-mix): the one-tick register transition P — a priori a Markov/GKLS transition rather than a deterministic map, since the ZS-F31 register clock is a GKLS clock — is *irreducible* and *doubly stochastic* on the $Q = 11$ slots (double

stochasticity = unitality of the tick channel on the diagonal: the tick preserves the counting measure in both directions). **(H-cycle)**: the deterministic special case — P is a transitive permutation σ (a single 11-cycle); $(H\text{-cycle}) \Rightarrow (H\text{-mix})$. Theorem T1' needs only $(H\text{-mix})$; the depth selection T3 additionally needs the exact period Q , which $(H\text{-cycle})$ supplies and $(H\text{-mix})$ alone does not. Two remarks discipline this condition. First, $Q = 11$ is prime, so a transitive cyclic action is automatically regular (Lemma F38.L1(ii)); there is no intermediate block structure to choose. Second, the alternative that the tick preserves the sector decomposition $2 + 3 + 6$ is not vacuous — deterministically it is the orbit type $(2, 3, 6)$, and in the Markov case the reducible chain with three closed classes; in both cases the stationary measures form the two-parameter sector-weighted family (Lemma F38.L1(iii); check PF3), which ZS-M44's exact arrowhead inverse-eigenvalue solve excluded to 0.45%. Within this paper's frame, the corpus's own exact solve is therefore **primitive-mixing evidence for (H-mix)** (Proposition F38.P1, CONFIRMED-NUMERICALLY as support, not proof) — broader than, and containing, the v1.0 permutation reading. Physically, transitivity is the statement that Z-Spin mediation genuinely mixes the X, Z, Y slots — consistent with the ZS-F37 mediation-vertex theorem (all three vertices survive with exact multiplicities $(2, 2, 2)$) — but the step-level permutation is not derived from the action here.

§4. Theorem F38.T1' and Corollary F38.T1 — Primitive Clock Ergodicity and Clock-Average Democracy

Theorem F38.T1' (Primitive Clock Ergodicity). *Assume $(H\text{-CLK})$ and $(H\text{-mix})$. Let P be the one-tick register transition, irreducible and doubly stochastic on the Q slots. Then: (i) the unique stationary probability measure of P is the uniform measure $\rho_Q = I_Q/Q$ — the uniform measure is stationary for every doubly stochastic P , and irreducibility makes the stationary measure unique (Perron–Frobenius); (ii) the Cesàro time average of every slot observable converges to the democratic average, and if P is additionally primitive (aperiodic) the convergence is direct, $P^n \rightarrow J/Q$; (iii) consequently, for every normalized slot vector $|s\rangle$ — equivalently every slot projector P_s — $\text{Tr}(\rho_Q P_s) = 1/Q$, and the ZS-M44 slot-normalized register-trace coupling is $\kappa^2 = A/Q$ with the democratic density ρ_Q obtained as the clock average, not asserted. **Status: mathematics IMPORTED-PROVEN (Perron–Frobenius; Birkhoff–von Neumann) + PROVEN in the finite instance (Appendix B.1; machine checks PF1–PF4, TR1); physical instance DERIVED-CONDITIONAL on $(H\text{-CLK}) \wedge (H\text{-mix})$.** *Review note (of record):* the v1.0 wording $\kappa^2 = A\langle a | \rho_Q | a \rangle = A/Q$ for an arbitrary slot observable a was over-broad — for general a , $\langle a | \rho_Q | a \rangle = \|a\|^2/Q$; the corrected statement (iii) uses normalized slot vectors/projectors, matching ZS-M44's actual normalized-trace bridge.*

Corollary F38.T1 (Clock-Average Democracy; the deterministic case). *Under $(H\text{-CLK})$ and $(H\text{-cycle})$ — $P = \sigma$ a transitive permutation, the extreme irreducible-periodic member of $(H\text{-mix})$ — statements (i)–(iii) hold with the Cesàro average realized exactly as the Birkhoff average over one clock period, $(1/Q) \sum_{k=0}^{Q-1} a(\sigma^k s) = \langle a \rangle_{I_Q/Q}$ for every starting slot s (checks UE1, UE4, PF4). [The v1.0 Theorem F38.T1, demoted to a corollary of T1'; note a permutation is irreducible with period Q but never primitive, so the deterministic case is the periodic extreme of $(H\text{-mix})$, not the primitive one.]*

Lemma F38.L1 (invariant simplex; prime rigidity). *For any permutation σ of a finite slot set: (i) a measure is σ -invariant iff it is constant on each orbit, so the invariant-measure simplex has dimension*

(#orbits – 1) (machine check UE6, five random permutations, exact); (ii) for $Q = 11$ prime, every nontrivial power of an 11-cycle is again an 11-cycle — the $\mathbb{Z}_{11}$ action is regular (UE3); (iii) the sector-preserving orbit type (2, 3, 6) has a 2-parameter invariant family containing the sector-weighted measures (UE2, UE5). [PROVEN]

Proposition F38.P1 (the M44 solve as primitive-mixing evidence). *The democratic outcome of the ZS-M44 exact arrowhead solve (couplings equal to 0.45%, decisively excluding the sector-weighted ratio $\varepsilon_X/\varepsilon_Y = 0.865$) is the unique outcome compatible with the irreducible branch of Theorem F38.T1' — deterministic or Markov — and is incompatible with a generic point of the reducible (sector-preserving) branch, whose stationary simplex is 2-dimensional (checks UE2, PF3).*

[CONFIRMED-NUMERICALLY as support for (H-mix); not a derivation of it.] The two ZS-M44 benchmark measures (mode-count vs metric) are hereby given a dynamical discriminator: an irreducible unital clock forces mode-count democracy as its time average — and the evidence class widens from “transitive permutation” to “primitive mixing,” the physically natural class for a GKLS clock.

Distinct-objects guard (DS2). The ZS-F37 two-leg transport weight $\omega = (4, 9, 36)/49$ is a Plancherel corner-transport weight; $\rho_Q = I_Q/Q$ is a state/trace density. They are different objects attached to the same register, and this paper claims no relation between them beyond coexistence; conflating them would contradict both ZS-M44 and ZS-F37. [GUARD; NON-CLAIM NC-F38.2]

§5. Theorem F38.T2 — the Q-Unit Inclusion Chain

Theorem F38.T2. *Assume KH1–KH4 (the ZS-M46 Suspension-Admissibility bundle), so that (M, U, Ω) is the positive-energy standard pair of the Z-Spin seam. Define $M_k := U(k) M U(-k)$ for $k = 0, 1, \dots, Q$. Then: (i) $M_0 \supset M_1 \supset \dots \supset M_Q$ is a nested chain, and each consecutive pair $M_{k+1} \subset M_k$ with the vector Ω is a half-sided modular inclusion (Borchers; Wiesbrock); (ii) the canonical endomorphism $\Gamma = J_N J_M = U(2)$ of ZS-M47 Theorem M47 (I) steps the chain by two, so the even subchain $M_0 \supset M_2 \supset \dots$ is the Jones–Longo tunnel; (iii) the chain order is characterized by inclusion of the spectral subspaces of the positive generators (Koot 2025), realized exactly in the Hardy model H^2 by inner-function divisibility: $M_k \leftrightarrow z^k H^2$, and $z^j \mid z^k$ iff $j \leq k$. Status: (i) IMPORTED-PROVEN (Borchers 1992; Wiesbrock 1993) + DERIVED bookkeeping; (ii) DERIVED-BY-INHERITANCE (ZS-M47, model-fixed conditions inherited); (iii) IMPORTED-PROVEN (Koot, arXiv:2503.18036) with exact model checks CH1–CH4.*

Lemma F38.L2 (standard-subspace bridge for the Koot criterion). *Let $K_k := \text{closure}((M_k)_{sa} \Omega)$ be the standard real subspace associated with (M_k, Ω) in standard form. Koot’s theorem is stated for half-sided modular inclusions of standard subspaces; it is applied here to the K_k -chain. Under KH1–KH4 the ZS-M46 seam pair lives in the Fock realization (ZS-M46 Theorem A), where the second-quantization functor $K \mapsto A(K)$ — Araki’s lattice of von Neumann algebras of the free field — sends standard-subspace inclusions to algebra inclusions and preserves the modular data; the algebra chain $M_{k+1} \subset M_k$ therefore inherits the spectral-subspace order criterion from the K_k -chain. [IMPORTED-PROVEN (Araki; Koot) + DERIVED bookkeeping; conditional only on KH1–KH4, already in T2’s hypothesis.] Review note (of record): v1.0 applied the criterion to the algebra chain directly; this lemma makes the standard-subspace domain of the imported theorem explicit.*

The chain is the finite object that the depth selector needs: a *ruler* of Q unit steps inside the seam inclusion, each step a genuine HSMI, ordered by a proven spectral criterion. Nothing in T2 selects Q over any other integer; the selection is the business of T3, and it comes from the register, not from the chain.

§6. Theorem F38.T3 — Conditional Depth Selection, $t^* = Q$

Orientation convention (fixed before the theorem, per ZS-A31 discipline). For the +half-sided inclusion the Borchers commutation reads $\Delta_M^{(-it)} U(a) \Delta_M^{(-it)} = U(e^{(-2\pi t)} a)$: forward modular time *contracts* translations (actor orientation). The reverse flow dilates them (stage orientation): $\Delta_M^{(-it)} U(a) \Delta_M^{(it)} = U(e^{(+2\pi t)} a)$. Both orientations are the same theorem; the 2π is Borchers–Wiesbrock-forced in either reading. The Planck convention is the reduced $\bar{M}_P$ (Locked Inputs), fixed before any comparison; the unreduced convention would shift the depth by $\approx 1/4$ unit (guard BW3) and is admissible only if a $Z^{(1/4)}$ seam operator is separately proven (ZS-A31 rule, inherited).

Definition F38.D3 (Evaluation condition, (H-eval)) [OPEN; decomposed in v1.1]. The emergent Kähler/metric length $\ell_K = M_K^{(-1)}$ is the seam-step length read on the stage-oriented branch of the modular flow at one full register-clock period. This is decomposed into three separately falsifiable sub-gates: **(H-eval-N) Normalization** — at $t = 0$ the unit seam step corresponds to the reduced Planck length ℓ_P ; **(H-eval-F) Functoriality** — the modular dilation parameter maps to physical length scaling; this sub-gate is carried by the emergent-metric structure, i.e. by the ZS-M47 NDC3 modular-Dirac operator (Theorem M47 (IV)), which itself inherits the ZS-M46 parent-injectivity condition through CRT-1; **(H-eval-B) Branch choice** — the stage-oriented branch evaluates length/time, the actor-oriented branch energy/mass. $(H\text{-eval}) \equiv (H\text{-eval-N}) \wedge (H\text{-eval-F}) \wedge (H\text{-eval-B})$.

Registry reconciliation (of record). ZS-A32 v1.0 §1.1 and §4.2 cite the T3 registry as $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4, \text{parent injectivity, NDC3}\}$. Under the decomposition above, “parent injectivity + NDC3” is precisely the (H-eval-F) inheritance chain spelled out, and the normalization/branch components were implicit in v1.0’s one-line (H-eval). **The two registries are identical**; this paragraph resolves the wording divergence of record, adding and removing no dependency on either side. The precise mathematical target that would discharge $(H\text{-eval-N}) \wedge (H\text{-eval-F})$ — the modular length functor E_{len} — is stated in Appendix D and scoped to ZS-M48/ZS-F39.

Theorem F38.T3. *Assume (H-CLK), (H-cycle), KH1–KH4, and (H-eval). Then: (i) one register-clock period is exactly Q units of seam modular time ((H-CLK) fixes the unit; (H-cycle) fixes the period); (ii) by the Borchers commutation, the unit seam step transported through one period on the stage-oriented branch is $U(e^{(+2\pi Q)})$ — the step has dilated by $e^{(2\pi Q)}$; (iii) hence $\ell_K = \ell_P e^{(2\pi Q)}$, equivalently*

$$*M_K / \bar{M}_P = e^{(-2\pi Q)} , \text{ i.e. } t = Q . **$$

Status: DERIVED-CONDITIONAL on the four-item registry $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4, (H\text{-eval})\}$. This converts the depth selector from NON-CLAIM (re-registered in ZS-M46, ZS-M47, and ZS-F37) and HYPOTHESIS-weak 30% (ZS-A31 §6.3, pre-registered) to a theorem with a short, named,

falsifiable registry. Numerically $e^{(-2\pi \cdot 11)} = 9.632 \times 10^{-31}$ (BW1); the observed effective ratio $M_{\text{eff}}/\bar{M}_P = 1.018 \times 10^{-30}$ (one firewalled observation, ZS-A31) then *infers* $C_{UV} = (M_{\text{eff}}/(\bar{M}_P e^{(-2\pi Q)})) \approx 1.25$, inside the a-priori band $[1/4, 4]$ (DS1) — an inference, not a fit: per the inherited A31 rule, only a *computed* C_{UV} may account for the 5.4% gap, and the gap must never be absorbed into **Q**.

What T3 does not do. It does not close B3. Closing B3 to DERIVED requires (α) the physical discharge of (H-CLK) — the C_{int} / CRT-4a computation, for which ZS-F37 Appendix D already specifies the finite protocol $\varepsilon_{C_{\text{int}}}$, now finite-decidable *given* the T2 chain — and (β) the full parent IPI computation of C_{UV} (ZS-F36 programme (i), untouched). The ZS-A17 / A27 / A28 no-go theorems are also untouched: T3 relates two scales; it does not generate a scale from (**A**, **Q**) alone. [**NON-CLAIM NC-F38.1, NC-F38.3**]

Corollary F38.C1 (the Planck pivot; statement only, handover). *The two orientations of the same commutation make the Planck scale a pivot: on the actor branch the unit step contracts (energies fall from the Planck maximum toward the wave floor M_K); on the stage branch it dilates (lengths and times grow from the Planck minimum toward cosmological maxima). Quantitatively, with the Friedmann relation $H^2 \propto \rho/\bar{M}_P^2$ (standard, external), the stage-side maxima obey $t_U \approx \bar{t}_P e^{(4\pi Q)} \cdot O(1)$ and the vacuum density $\rho_A/\bar{M}_P^4 \approx e^{(-8\pi Q)} \cdot O(1)$ — one depth number, four corners. [Statement only; the quantitative ladder, its observational comparison, and the executed Monte Carlo are scoped to ZS-A32 (NC-F38.5).]*

§7. Effect on the corpus ledger and cross-version consistency

Table 7.1 — The ZS-F36 v2.1 three-programme ledger, before and after ZS-F38.

Programme	Before (F36 v2.1 / M44 v1.6 / A31 v1.5)	After ZS-F38 v1.0
(i) C_{UV} full parent IPI	OPEN (separate programme)	OPEN, unchanged; but the C_{int} half of the clock gate is finite-decidable given the T2 chain (F37 App. D protocol; $\varepsilon_{C_{\text{int}}}$)
(ii) register measure $\rho_Q = I_Q/Q$	ASSERTED (M44: DERIVED-CONDITIONAL on register-trace normalization; density not action-derived)	DERIVED-CONDITIONAL on (H-CLK) $\wedge$ (H-mix) via Theorem F38.T1'; deterministic (H-cycle) case retained as Corollary F38.T1; M44's 0.45% democratic solve becomes evidence for the irreducible/unital branch.
(iii) $M_K/\bar{M}_P$ hierarchy (B3)	OPEN debt; candidate $e^{(-2\pi Q)}$ HYPOTHESIS-weak 30% (A31 §6.3); $t^* = Q$ NON-CLAIM (M46/M47/F37)	$t^* = Q$ DERIVED-CONDITIONAL on {(H-CLK), (H-cycle), KH1–KH4, (H-eval-N/F/B)} (Theorem F38.T3); statistical leg closed in ZS-A32 ($p_{\text{single}} = 0.50\%$, corner (W) HYPOTHESIS-strong); B3 = two-gate debt {(H-CLK), C_{UV} }

Cross-version consistency (protocol 3.2). ZS-F38 changes no upstream numerical value and reverses no upstream status. Consumed verbatim: ZS-M46 v1.5 (M46.3A unit translation; KH1–KH4; CRT-4a as OPEN), ZS-M47 v2.0 (Theorem M47 (I): $\Gamma = U(2)$; the tunnel; the handover), ZS-F37 v1.3 (App. C.5 gate wording; App. D $\varepsilon_{C_{\text{int}}}$; the (H- Σ_2)/R2b tie; $\omega = (4,9,36)/49$ as a distinct object), ZS-M44 v1.6 (the conditional status of $\kappa^2 = A/Q$; the arrowhead 0.45% result; the §10.4 artifact lesson), ZS-F36 v2.1 (the three-programme terminus; $1260/4807 = 36A/Q$), ZS-A31 v1.5 (the reduced- $\bar{M}_P$ convention; the pre-registered candidate and its 5.4%/ C_{UV} reading; the firewalled $M_{\text{eff}} = 2.48$ meV), the ZS-A17/A27/A28 no-gos, and — new in v1.1 — ZS-A32 v1.0 (its executed MC values $p_{\text{single}} = 0.500\%/0.499\%$ and $p_{\text{universe}} = 28.76\%$, its corner table, and its model-level $\varepsilon_{C_{\text{int}}} = 0$ on this

paper's T2 chain are consumed as reported and reproduced here as checks MC5–MC7). Two reconciliations of record are made in v1.1, both wording-level with both sources preserved: the T3 registry (§6) and the Appendix C statistic hierarchy (§10, with the v1.0 formula-count reading retained as SECONDARY and passing at $2/88 = 2.27\%$). The locked dynamical digits (z, λ, μ, θ) are re-verified in-suite (KO1–KO2), and the Abel increment and branch period match ZS-M47 GEO2–GEO3 (KO3–KO4). No observational claim beyond ZS-A31's single firewalled regression is introduced (protocol 3.3).

§8. Falsification gates

F-F38.1 (clock speed). If CRT-4a is resolved with a non-central cocycle, or the seam and register clocks are shown to run at unequal speeds, (H-CLK) fails and T1(iii)/T3 collapse to their prior statuses (M44 asserted; A31 HYPOTHESIS-weak).

F-F38.2 (sector-preserving clock). If the tick transition is proven sector-preserving (deterministic orbit type (2,3,6) or a reducible Markov chain), (H-mix) and (H-cycle) fail; democracy loses its clock derivation — and the M44 0.45% democratic solve then stands unexplained within this frame (a registered tension, not a corpus contradiction).

F-F38.3 (depth mismatch). If the computed full-IPI C_{UV} falls outside $[1/4, 4]$, or a sharper M_{eff} pins $M_K/\bar{M}_P$ away from $e^{(-2\pi Q)}$ beyond the C_{UV} band, Theorem F38.T3's registry is falsified in its (H-eval) or (H-CLK) item (inherits ZS-A31's falsification wording).

F-F38.4 (admissibility). If any of KH1–KH4 fails, the standard pair and hence the T2 chain and T3 collapse (inherited from ZS-M46).

F-F38.5 (Monte Carlo; executed). ZS-A32 v1.0 executed the Appendix C design: PRIMARY $p_{single} = 0.50\% \leq 5\%$ (PASS), SECONDARY formula-count $2/88 = 2.27\% \leq 5\%$ (PASS), look-elsewhere 28.8% (audit, of record). The gate now guards re-executions: any future re-run under changed inputs returning both decision candidates $> 5\%$ withholds the corner per the corpus rule (ZS-M47 CY4 precedent); any demonstration that §10/Appendix C mis-states the registration order (target ZS-A31-first) voids the primary and falls back to the formula-count reading.

F-F38.6 (chain order). If the Koot spectral criterion fails for the concrete Z-Spin realization of the M_k chain (non-inner relative position), T2(iii) fails and the ruler loses its order theorem.

F-F38.7 (convention). If a $Z^{(1/4)}$ seam operator is proven from the action, the unreduced-convention depth $Q + 1/4$ must be re-examined against the reduced-convention pre-registration (inherited A31 rule; guard BW3).

F-F38.8 (unitality/irreducibility). If the ZS-F31 tick channel is proven non-unital on the diagonal (not doubly stochastic) or reducible, Theorem F38.T1' loses its physical instance; the mathematics stands, the register application falls.

§9. Non-claims

NC-F38.1. No closure of B3 (the H_0/M_P hierarchy): (H-CLK) is undischarged and C_{UV} uncomputed. **NC-F38.2.** No identification of the Plancherel transport weight $\omega = (4,9,36)/49$ with the state density I_Q/Q (distinct objects; guard DS2). **NC-F38.3.** No derivation of any dimensional scale from (A, Q) alone (ZS-A17/A27/A28 no-gos untouched). **NC-F38.4.** The Monte Carlo execution is ZS-A32 v1.0's, consumed as reported and reproduced as checks (MC5–MC7); this paper adjudicates nothing beyond the hierarchy of record. **NC-F38.7.** The modular length functor E_{len} (Appendix D) is a proof *target*, not a construction; its delivery is scoped to ZS-M48/ZS-F39. **NC-F38.5.** The quantitative four-corner extremal ladder (t_U , R_U , ρ_Λ , and the telomere reading $n_{Hubble} = 4QA$) is scoped to ZS-A32. **NC-F38.6.** No claim on the Clay-form Yang–Mills mass gap or any Millennium problem (ZS-F18 discipline inherited).

§10. Anti-numerology and the single-hypothesis structure

The paper's only numerical frontier statement is the inherited, already-pre-registered candidate $M_K/\bar{M}_P = e^{(-2\pi Q)}$ (ZS-A31 v1.2 §6.3). Theorem F38.T3 adds *mechanism structure* (a registry), not a new number; the stage-side corners of Corollary F38.C1 are Friedmann consequences of the same single number — **no additional fit, no multiple-testing burden**.

The statistical hierarchy of record (v1.1 correction; verdict-neutral). The v1.0 Appendix C froze the universe ($|U| = 88$), the band $((1/4) \ln 4)$, the single target $((c, q) = (2, 11))$, and the rule ($p \leq 5\%$), but its prose defined p as the *formula-count* fraction, while ZS-A32 v1.0 — correctly reflecting the registration logic — executed the *single-target* statistic as primary. v1.1 fixes the hierarchy of record: **PRIMARY** $p_{single} = 2 \cdot \text{band} / \ln(\bar{M}_P/H_0) = 0.500\%$ (the target $e^{(-2\pi Q)}$ was registered in ZS-A31 v1.2 *before* this universe existed, so the single-target chance rate is the statistic the registration earns; the null range is the widest hierarchy realized in nature, built from observed quantities only); **SECONDARY** $p_{\text{formula-count}} = 2/88 = 2.27\%$ (the v1.0 literal reading — exactly two members lie in the band of the observed corner, and they are the disclosed degenerate pair $2\pi \cdot 11 = \pi \cdot 22$; checks MC4–MC5); **SECONDARY (look-elsewhere audit)** $p_{\text{universe}} = 28.8\%$ (interval-union coverage — the measured size of the numerological temptation a non-registered search would have faced; check MC7); **REPORTED ONLY** $p_{\text{sharp}} = 2|\Delta|/W = 0.079\%$. The correction is **verdict-neutral**: both decision candidates pass at $\leq 5\%$ (guard MC3'), so the change relabels, it does not rescue. Publishing p_{universe} alongside prevents the converse abuse. The ZS-M44 §10.4 lesson (an exact-looking ratio mimicking a structural constant) and the ZS-M47 CY4 precedent (execute, and withhold when p says withhold) remain the governing discipline.

§11. Verification suite

Table 11.1 — `zs_f38_verify_v1_1.py`: 32/32 exact/numerical PASS + 3/3 guards (SymPy exact + mpmath 60-digit + NumPy).

Block	Checks	Content
PF/TR (T1')	PF1–PF4, TR1	irreducible doubly stochastic $P = (C + C^2)/2$: unique stationary $= I_Q/Q$ (exact rational); primitivity: $P^{15} > 0$ and $P^{1000} \rightarrow J/Q$ ($< 10^{-12}$); reducible sector chain:

		stationary dimension 3; Cesàro average of the 11-cycle = J/Q exactly; $\text{Tr}(\rho_Q P_s) = 1/Q$ for all 11 slot projectors
UE (T1/L1)	UE1–UE6	transitive 11-cycle: $\dim(\text{inv})=1$, measure = I_Q/Q ; sector alternative $\dim=3$; Z_{11} regularity; Birkhoff period average (all 11 starts, exact rationals); sector-weighted invariant $\neq$ uniform; simplex dimension = #orbits (5 random permutations)
KO (inherited data)	KO1–KO4	z, μ, θ to locked 10 digits; Koenigs ratio $\chi(f_z) = \lambda\chi(z)$ and Abel increment = 1 at 12 orbit points ($<10^{-12}$); branch period $2\pi i/\log \lambda^* = 2.7743 - 0.1410i$ (M47 GEO3)
CH (T2)	CH1–CH4	Hardy/ideal model: chain order by divisibility (all $0 \leq j \leq k \leq Q$); $\Gamma = U(2)$ even subchain; unit-shift bookkeeping; Koot order criterion iff $j \leq k$
BW (T3)	BW1–BW2 + guard BW3	$e^{(-2\pi)}$ and Q -fold composition = 9.632×10^{-31} ; $t_{\text{obs}}(\text{reduced}) = 10.991$, $ t_{\text{obs}} - Q = 0.0089$ (reproduces A31); unreduced $\approx Q + 1/4$ convention guard
DS (consistency)	DS1, DS3–DS4 + guard DS2	$C_{UV} = 1.25 \in [1/4, 4]$; $\kappa^2 = 35/4807$, $36A/Q = 1260/4807$; $\ln(\bar{M}_P/M_{\text{eff}}) = 69.059$ vs $2\pi Q = 69.115$ (0.081%, consistency only); distinct-measures guard
MC (hierarchy of record)	MC1–MC2, MC4–MC7 + guard MC3'	$ U = 88$ and target membership; band $(1/4) \ln 4$; degenerate pair $2\pi \cdot 11 = \pi \cdot 22$ exact; SECONDARY formula-count $2/88 = 2.27\%$; PRIMARY $p_{\text{single}} = 0.500\%$ (A32 value reproduced); look-elsewhere 28.8% (matches A32 28.76%); verdict-neutrality guard
PR (locked echoes)	PR1–PR2	$2 + 3 + 6 = 11 = Q$; (A, Q, $\dim Z$) echo

The suite certifies the mathematics of T1/L1, the model form of T2, the arithmetic of T3, and the inherited dynamical digits. It does **not** certify (H-CLK), (H-cycle), (H-eval), or KH1–KH4 — these are named conditions, exactly as ZS-M46/M47 handle their imported-theorem registries.

§12. Conclusion and handover to ZS-A32

One identity was hiding behind five open lines. Named and bundled as (H-CLK), it lets Perron–Frobenius ergodic theory derive the democratic register measure (T1'), lets the Borchers–Wiesbrock chain provide a proven Q -step ruler with an explicit standard-subspace bridge (T2, L2), and lets one commutation relation select the modular depth $t = Q$ (T3) — *all conditionally, with every condition named in the theorem's first sentence. Positioning (of record). This paper's external value does not rest on the number $e^{(-2\pi Q)}$, which remains a theory-laden conditional claim; it rests on three transferable structures: (1) a finite register clock selecting the democratic trace by unique ergodicity (T1' — a statement about any unital irreducible finite clock); (2) an HSMI/standard-pair chain used as a finite Q -depth ruler with a proven order criterion (T2 + L2); and (3) a worked pre-registration protocol converting a loose depth numerology into a named conditional theorem, a falsification registry, and an executed, hierarchy-fixed statistical verdict (§10, Appendix C, ZS-A32). The corpus pattern holds: internal iteration converges to honesty, and the honest statement here is a compression, not a closure**. The keystone that remains is the physical discharge of (H-CLK) — the $C_{\text{int}} / \text{CRT-4a}$ computation, whose model level ZS-A32 already executed at $\epsilon_{C_{\text{int}}} = 0$ exactly on this paper's T2 chain — and the C_{UV} IPI.

Table 12.1 — Handover to ZS-A32.

Item	Deliverable in ZS-A32	Source here
Four-corner extremal ladder	quantitative table: $M_{\text{eff}}, \rho_A, t_U, R_U, H_0, n_{\text{Hubble}} = 4QA$ reading; Friedmann-forced squares made explicit	Corollary F38.C1
Anti-numerology MC	EXECUTED in ZS-A32 v1.0: $p_{\text{single}} = 0.50\%$, formula-count 2.27%, look-elsewhere 28.8%; corner (W) promoted HYPOTHESIS-strong	Appendix C (hierarchy of record)

(H-2D), P2 envelope, (★_w), (TT)	per the ZS-M47 §12.3 handover rows (untouched here)	ZS-M47
ϵ_C_int execution	model level EXECUTED in ZS-A32 ($\epsilon = 0$ exactly on the T2 chain); physical level OPEN behind (H-CLK)/(H-Σ2) → ZS-M48/ZS-F39	§6, T2
Status promotion	corner (W) now HYPOTHESIS-strong // DERIVED-CONDITIONAL (A32); next: (H-CLK) discharge shortens T1'/T3; E_len delivery discharges (H-eval-N/F) → ZS-M48/ZS-F39	§10, Appendix D

§13. Acknowledgements & Code Availability

This work consolidates internal Z-Spin Collaboration deep-exploration notes following ZS-F37 v1.3 and ZS-A31 v1.5, and executes the gate pre-registered in ZS-F37 Appendix C.5. The verification script `zs_f38_verify_v1_1.py` (SymPy + mpmath + NumPy) reproduces all 32 exact/numerical checks and 3 guards; the ergodicity blocks are exact rational linear algebra plus a numerical Perron–Frobenius power check, the Koenigs block runs at 60-digit precision, the chain block is exact polynomial-ideal arithmetic, and the MC block reproduces the ZS-A32 executed statistics (0.500%, $2/88 = 2.27\%$, 28.8%) from the frozen design. This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility for all content.

Appendix A — Deep-exploration (issue-tree) record [pre-registered, then executed]

Step 0 (long list, 7). (1) axiomatize the four-corner extremal duality directly; (2) classify the depth lattice $\pi\mathbb{Q}(\mathbf{A}, \mathbf{Q})$ of inter-extremal exponents; (3) the Register Clock Identity — one gate behind programmes (ii)/(iii) and the CRT-4a/C_int gates; (4) compute C_UV full-1PI directly; (5) build the genuine Regge Hessian B_Whinge directly; (6) the holographic information-budget reading $S_{dS} \approx e^{(8\pi\mathbf{Q})}$; (7) a unification principle for the A-family ($\pi/\mathbf{A}$, $35\pi/3$) and Q-family ($2\pi\mathbf{Q}$) exponents. **Dropped:** (1) — axiomatizing the duality first would be numerology-first, conflicting with the A31 pre-registration discipline (demoted to Corollary C1); (4) and (5) — genuine computations, not abstractions; routed to protocols (F37 App. D; M44 §6.3 successor), kept OPEN.

Step 1 (MECE issue list, 4, by impact). **I1** = the Register Clock Identity {3} (keystone); **I2** = depth lattice + four corners {2, 7, residue of 1}; **I3** = finite computation gates {4, 5} (supporting, I1-dependent); **I4** = holographic cross-check {6} (packaging axis). Nothing further dropped; I4 carried as NON-CLAIM-grade context.

Step 2 (tree). $I1 \rightarrow I2 \rightarrow I4$, with I3 parallel under I1: I1a seam unit translation; I1b integer depth quantization; I1c the identity itself; I1d democratic measure; I1e depth selection; I2-W/T/L/P the corners; I2-n the telomere reading.

Step 3 (statuses). I1a PROVEN (M46.3A; re-verified KO3). I1b IMPORTED-PROVEN + DERIVED (Jones–Longo; M47 (I); CH1–CH4). I1c OPEN — the single keystone ((H-CLK); condition reuse). I1d DERIVED-CONDITIONAL on I1c + (H-cycle) (Theorem T1). I1e DERIVED-CONDITIONAL on I1c + (H-cycle) + KH1–KH4 + (H-eval) (Theorem T3). I2-W: the A31 candidate, now theorem-backed (T3); I2-T/L DERIVED-CONDITIONAL on I2-W + Friedmann (external, standard); I2-P honestly A-family

(F24-7; S7), not forced by the clock — registered separately; I2-n the same single hypothesis re-expressed (n = 4QA = 1540/437). I3 OPEN by design (protocols specified). I4 OBSERVATION/NON-CLAIM.

Step 4 (convergence). Cycle 1 assigned statuses to 10 nodes. Cycle 2 changed 2: (H-cycle) was added as a named condition on I1e after the sector-preserving alternative was found non-vacuous (orbit type (2,3,6) — the decisive honesty item of this exploration), and I2-P was moved off the Q-ladder to the A-family. Cycle 3 changed 0. Change counts $10 \rightarrow 2 \rightarrow 0$, strictly decreasing $\Rightarrow$ **converged**; MECE retained.

Step 5 (scoring). Converged + corpus-non-colliding (A31 pre-registration, A17/A27/A28 no-gos, F37 C.5 scope, M44 conditional statuses all respected) + anti-numerology MC **pre-registered but not executed** $\Rightarrow$ per the corpus rubric the numerical depth statement remains at its inherited pre-registered level until the ZS-A32 execution (registrable HYPOTHESIS-strong upon $p \leq 5\%$); the structural value is the compression of three OPEN programmes and two OPEN gates into one bundled identification plus one computation, with the register measure gaining its first derivation route.

v1.1 addendum — review-integration exploration (Steps 0–5, executed for this version). *Step 0 (long list, 6):* (R1) MC statistic reconciliation with ZS-A32; (R2) registry unification via (H-eval) decomposition; (R3) T1 narrowing + Perron–Frobenius generalization; (R4) Koot standard-subspace bridge; (R5) title/positioning correction; (R6) full E_len functor construction. *Dropped:* (R6) as a construction — too heavy for a revision; retained as a stated proof target (Appendix D), scoped to ZS-M48/ZS-F39. *Step 1 (MECE, 5 by impact):* I1 = R1 (protocol integrity — the only item that could void a verdict); I2 = R2 (dependency-graph integrity); I3 = R3 (mathematical attack surface + strengthening); I4 = R4 (external readability of the imported theorem); I5 = R5 (claim-scope hygiene). *Step 2:* I1 $\rightarrow$ I2; I3 $\rightarrow$ I4; I5 global. *Step 3:* I1 resolved verdict-neutral (both decision candidates $\leq 5\%$; hierarchy justified by A31-first registration; duplicate $2\pi \cdot 11 = \pi \cdot 22$ disclosed) [DERIVED, arithmetic + protocol]; I2 resolved by (H-eval-N/F/B) with the A32 wording identified as (H-eval-F) spelled out [reconciliation of record]; I3: T1' IMPORTED-PROVEN + PROVEN (PF1–PF4), κ^2 narrowed to slot projectors (TR1), evidence class widened to primitive mixing; I4: Lemma L2 via the Araki second-quantization functor under KH1–KH4 [IMPORTED-PROVEN + DERIVED]; I5: “executing” $\rightarrow$ “registering” throughout. *Step 4:* cycle 1 changed 5 items; cycle 2 changed 1 — the T1' condition was corrected from the review’s “primitive” to the exact “irreducible” for uniqueness, with primitivity retained as the strengthening for direct convergence, since a permutation is never primitive and the v1.0 case would otherwise have fallen outside its own generalization; cycle 3 changed 0. Sequence $5 \rightarrow 1 \rightarrow 0$: converged. *Step 5:* every review item implemented with zero conclusion, value, or verdict changes; the one substantive mathematical fix (irreducible vs primitive) was caught by this exploration’s own re-traversal — the protocol working as designed.

Appendix B — Proofs

B.1' (Theorem F38.T1'). For doubly stochastic P the uniform vector is stationary (column sums 1). If P is irreducible, Perron–Frobenius gives a one-dimensional stationary space, so $\rho_Q = I_Q/Q$ uniquely [(i); check PF1]. The Markov-chain ergodic theorem gives Cesàro convergence of time averages to the stationary average for irreducible chains, and direct convergence $P^n \rightarrow \mathbb{1} \rho_Q^T = J/Q$ when P is additionally primitive [(ii); checks PF2, PF4]. For a normalized slot vector, $\text{Tr}(\rho_Q P_s) = 1/Q$ [(iii);

check TR1]; substituting into the ZS-M44 normalized-trace bridge gives $\kappa^2 = A/Q$. A transitive permutation is irreducible with period Q — never primitive — so Corollary T1 is the periodic extreme, its Cesàro average realized exactly by one period (check PF4). By Birkhoff–von Neumann, every doubly stochastic P is a convex combination of permutations: the deterministic case is the extreme-point skeleton of (H-mix). **B.1 (Lemma F38.L1 and Corollary F38.T1; the v1.0 proof, retained).** Let σ be a permutation of the finite slot set S , $|S| = Q$, and P_σ the induced operator on measures, $(P_\sigma \rho)(s) = \rho(\sigma^{-1} s)$. Then $P_\sigma \rho = \rho$ iff $\rho(\sigma^{-1} s) = \rho(s)$ for all s , i.e. iff ρ is constant on each $\langle \sigma \rangle$ -orbit; the invariant measures are the convex hull of the orbit-uniform measures, a simplex of dimension $(\# \text{orbits} - 1)$ [L1(i)]. If σ is transitive there is one orbit, so $\rho = I_Q/Q$ uniquely [T1(i)]. For Q prime, a transitive action of the cyclic group $\langle \sigma \rangle$ has trivial point stabilizers by orbit–stabilizer, hence is regular, and every nontrivial power of σ is again an 11-cycle [L1(ii)]. For orbit type (2, 3, 6) the simplex is 2-dimensional and contains all sector-weighted measures [L1(iii)]. Birkhoff over one period: since the orbit of any s visits each slot exactly once in Q steps, $(1/Q) \sum_{k=0}^{Q-1} a(\sigma^k s) = (1/Q) \sum_{s' \in S} a(s') = \langle a \rangle_{I_Q/Q}$ [T1(ii)]; substituting $\rho_Q = I_Q/Q$ into the ZS-M44 normalized-trace bridge gives $\kappa^2 = A/Q$ [T1(iii)].

B.2 (Theorem F38.T2). By the standard-pair axioms $U(a)MU(-a) \subset M$ for $a \geq 0$ (Borchers), so $M_k = U(k)MU(-k)$ is nested and each $M_{k+1} = U(1)M_k U(-1) \subset M_k$ is +half-sided with the same Ω ; Wiesbrock's theorem gives each pair the HSMI structure. $\Gamma = U(2)$ is ZS-M47 Theorem M47 (I), inherited with its conditions; iterating Γ gives the even subchain = the Jones–Longo tunnel. Koot's theorem (arXiv:2503.18036) characterizes $K_1 \subset K_2$ for two HSMIs in a common standard object by inclusion of spectral subspaces of the positive generators, with the (operator-valued) symmetric-inner-function normal form; in the Hardy model $U(k)$ is multiplication by the inner function z^k , spectral inclusion is divisibility, and $z^j \mid z^k$ iff $j \leq k$ (checks CH1–CH4, exact).

B.3 (Theorem F38.T3). (H-CLK) equates the register clock with the seam modular flow σ_t^M at equal speed (a central cocycle cannot rescale a flow), with the ZS-F31 GKLS normalization one slot per unit t . (H-cycle) makes the slot dynamics a single Q -cycle, so the register returns after exactly Q units of modular time — the period is Q . The Borchers commutation for the standard pair gives $\Delta_M^{\mp it} U(a) \Delta_M^{\pm it} = U(e^{\pm 2\pi t} a)$; on the stage-oriented branch at $t = Q$ the unit seam step $U(1)$ is carried to $U(e^{2\pi Q})$. (H-eval) identifies ℓ_K with that period-return step length in Planck units, $\ell_K = \ell_P e^{2\pi Q}$; inverting, $M_K/\bar{M}_P = e^{-2\pi Q}$, i.e. $t = Q^*$. Every named condition is used exactly once, and removing any one of them removes the conclusion.

Appendix C — The anti-numerology protocol: hierarchy of record (v1.1) and the ZS-A32 execution

Correction of record (v1.1). The v1.0 appendix froze the design — universe, band, target, rule — but its prose defined the decision statistic as the formula-count fraction, whereas the registration logic (target fixed in ZS-A31 v1.2 before this universe existed) makes the single-target chance rate primary, and ZS-A32 v1.0 executed it as such. This version fixes the hierarchy; the frozen design objects are unchanged, and the correction is verdict-neutral (guard MC3': both decision candidates $\leq 5\%$).

Target. $x_{\text{obs}} = \ln(\bar{M}_P/M_{\text{eff}})$, reduced convention, M_{eff} the single firewalled ZS-A31 regression (ZS-A32 value 2.476 meV: $x_{\text{obs}} = 69.061$, $|\Delta| = 0.054$). The registered exact target is the emergent-metric reading $M_K = \bar{M}_P e^{(-2\pi Q)}$, with $M_{\text{eff}} = C_{UV}^{(1/4)} M_K$.

Universe U ($|U| = 88$; **grammar fixed here, never to be extended post-hoc**). (a) $\exp(-c \cdot \pi \cdot q)$ with $c \in \{1/2, 1, 2, 4\}$ and $q \in \{2, 3, 6, 11, 22, 28, 33, 35, 49, 121\}$ — 40 members (the corpus-constant grid: dims, $Q, 2Q, |V_Y - F_Y|, 3Q, 35, 49, 121$); (b) A^k , $k = 1, \dots, 40$ — 40 members (the A-power family that ZS-A31 §6.3 demoted); (c) $\exp(-n \cdot \pi/A)$, $n = 1, \dots, 8$ — 8 members (the telomere family). The target is member $(c, q) = (2, 11)$ of family (a) (check MC1).

Tolerance band. $|E - x_{\text{obs}}| \leq (1/4) \ln 4 = 0.3466$, the a-priori C_{UV} band $[1/4, 4]$ read through $M_{\text{eff}} = C_{UV}^{(1/4)} M_K$ (check MC2). **Degeneracy disclosure.** The universe retains registered degenerates; in particular $\pi \cdot 22 = 2\pi \cdot 11$ exactly (check MC4), so the target value appears twice — deduplication would shrink coverage and flatter the verdict (the ZS-A32 rule, adopted).

The statistical hierarchy of record. PRIMARY: $p_{\text{single}} = 2 \cdot \text{band}/W$, $W = \ln(\bar{M}_P/H_0) = 138.68$ (the widest hierarchy realized in nature, observed quantities only); executed value 0.500% analytic, 0.499% at 10^6 draws (ZS-A32; reproduced as check MC6). SECONDARY: $p_{\text{formula-count}} = (\text{members of } U \text{ in the band of } x_{\text{obs}})/|U| = 2/88 = 2.27\%$ — the v1.0 literal reading, retained and passing; the two members are the degenerate pair above (check MC5). SECONDARY (look-elsewhere audit): $p_{\text{universe}} = \text{exact interval-union coverage} = 28.8\%$ (ZS-A32: 28.76%; reproduced as check MC7) — the measured size of the temptation a non-registered search would have faced; explicitly not a decision statistic, published to prevent the converse abuse. REPORTED ONLY: $p_{\text{sharp}} = 2|\Delta|/W = 0.079\%$. **Decision rule.** Adjudicate on the primary; the formula-count is the fallback if the registration order is ever impeached (F-F38.5). $p \leq 5\% \Rightarrow$ registrable HYPOTHESIS-strong (achieved: ZS-A32 promoted corner (W)); both candidates $> 5\% \Rightarrow$ withhold (ZS-M47 CY4 precedent). **Convention locks.** Reduced $\bar{M}_P$ only; no post-hoc absorption of the 5.4% gap into Q ; the unreduced $Q + 1/4$ reading admissible only under a proven $Z^{(1/4)}$ seam operator (guard BW3). **Why the single-target statistic is primary.** The target was pre-registered in ZS-A31 v1.2 before the universe was designed; the universe was built afterward precisely to measure look-elsewhere. Adjudicating on p_{universe} would penalize the hypothesis for a search that never happened; adjudicating on formula-count alone would let universe-design choices move the verdict. Both are of record.

Appendix D — The modular length functor E_{len} : the (H-eval-N/F) discharge target [proof target; scoped to ZS-M48/ZS-F39]

Target. Construct a functor $E_{\text{len}} : \text{StandardPair}_{Z\text{-Spin}} \rightarrow \mathbb{R}_+$ on the category of Z-Spin seam standard pairs (objects: (M, U, Ω) under KH1–KH4; morphisms: cocycle-covariant isomorphisms) such that $(N) E_{\text{len}}(U(a)) = a \cdot \zeta_P$ for $a \geq 0$, and $(F) E_{\text{len}}(\Delta^{(-it)} U(1) \Delta^{(it)}) = e^{(2\pi t)} \zeta_P$. Delivery of E_{len} discharges (H-eval-N) $\wedge$ (H-eval-F) and converts the T3 conclusion $\ell_K = \zeta_P e^{(2\pi Q)}$ from a physical evaluation condition into a functorial evaluation theorem; (H-eval-B) then reduces to the orientation bookkeeping fixed in §6, and the T3 registry shortens to $\{(H\text{-CLK}), (H\text{-cycle}), \text{KH1–KH4}\}$. The natural construction route is the ZS-M47 NDC3 modular-Dirac spectral geometry (the length element as the inverse Dirac scale; Connes distance on the compact dual), which is exactly why (H-eval-F) inherits

parent injectivity through CRT-1. **Status: stated target only (NC-F38.7); any claim of delivery belongs to ZS-M48/ZS-F39.**

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Version History

v1.1 (July 2026, review-integrated): Integrates a detailed external review of v1.0 together with the executed ZS-A32 v1.0; no theorem conclusion, numerical value, or epistemic verdict changes. **(1) Statistical hierarchy of record (the decisive item):** Appendix C corrected — the v1.0 prose defined the decision statistic as the formula-count fraction while ZS-A32 correctly executed the single-target statistic as primary (the target being ZS-A31-v1.2-registered before the universe existed); v1.1 fixes PRIMARY $p_{\text{single}} = 0.500\%$, SECONDARY $p_{\text{formula-count}} = 2/88 = 2.27\%$ (the v1.0 literal reading, retained and passing; the two in-band members are the disclosed degenerate pair $2\pi \cdot 11 = \pi \cdot 22$), look-elsewhere $p_{\text{universe}} = 28.8\%$, $p_{\text{sharp}} = 0.079\%$ reported — verdict-neutral under both decision candidates (guard MC3'). **(2) Registry reconciliation:** (H-eval) decomposed into (H-eval-N) normalization, (H-eval-F) functoriality (carried by NDC3, inheriting parent injectivity via CRT-1), and (H-eval-B) branch choice; the ZS-A32 §1.1 registry wording is identified of record as (H-eval-F) spelled out — the registries are identical. **(3) T1 corrected and strengthened:** the over-broad $\kappa^2 = A\langle a|p_Q|a\rangle$ narrowed to the slot-projector form $\text{Tr}(\rho_Q P_s) = 1/Q$ (check TR1); Theorem F38.T1' (Primitive Clock Ergodicity) added — an irreducible doubly stochastic (unital) tick has unique stationary measure I_Q/Q with Cesàro convergence, primitive $\Rightarrow$ direct convergence (Perron–Frobenius; Birkhoff–von Neumann; checks PF1–PF4) — under the new condition (H-mix), with the v1.0 theorem demoted to Corollary F38.T1 ((H-cycle) $\Rightarrow$ (H-mix)); the exploration's own re-traversal corrected the review's “primitive” to “irreducible” for uniqueness, since a permutation is never primitive. **(4) Koot bridge:** Lemma F38.L2 — the criterion applied to the standard subspaces $K_k = \text{closure}((M_k)_{sa} \Omega)$ and transported to the algebra chain by the Araki second-quantization functor under KH1–KH4. **(5) Scope hygiene:** “executing” $\rightarrow$ “registering” the ZS-F37 C.5 gate (subtitle, metadata, §1, §13); positioning paragraph added (§12: the paper's external value = the ergodicity theorem, the Q-depth ruler, and the worked pre-registration protocol — not the number). **(6) E_len proof target** stated (Appendix D; NC-F38.7), scoped to ZS-M48/ZS-F39. **(7) Gates:** F-F38.5 updated to the executed ZS-A32 values; F-F38.8 (unitality/irreducibility) added. Verification 23/23 $\rightarrow$ **32/32** + 3/3 guards (zs_f38_verify_v1_1.py: new PF1–PF4, TR1, MC4–MC7). Zero fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. (Consolidated from internal Z-Spin Collaboration deep-exploration notes following the external review of ZS-F38 v1.0 and ZS-A32 v1.0.)

v1.0 (July 2026): Initial public release. Executes the ZS-F37 Appendix C.5 pre-registered gate under the code ZS-F38: the Register Clock Identity (H-CLK) defined as condition reuse ($F37\ C.5 \wedge \text{CRT-4a}$, ZS-F31 GKLS normalization); Theorem F38.T1 (Clock-Average Democracy; unique ergodicity, mathematics PROVEN, physics DERIVED-CONDITIONAL on $(H\text{-CLK}) \wedge (H\text{-cycle})$); Lemma F38.L1 (invariant simplex; prime-Q rigidity; the sector-preserving (2,3,6) alternative identified with the ZS-M44-excluded sector-weighted family); Theorem F38.T2 (the Q-unit inclusion chain; Borchers/Wiesbrock + Jones–Longo + Koot, Hardy-model exact checks); Theorem F38.T3 (Conditional Depth Selection $t^* = Q$, DERIVED-CONDITIONAL on the four-item registry $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4, (H\text{-eval})\}$) — converting the ZS-A31 §6.3 HYPOTHESIS-weak candidate and the M46/M47/F37 NON-CLAIM into a theorem with a named registry, without closing B3; Corollary F38.C1 (Planck pivot, statement only, handover to ZS-A32); the pre-registered anti-numerology formula universe ($|U| = 88$, band $(1/4)\ln 4$, rule $p \leq 5\%$; execution scoped to ZS-A32). Verification 23/23 PASS + 3/3 guards (zs_f38_verify_v1_0.py). Zero fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

(Consolidated from internal Z-Spin Collaboration deep-exploration notes following ZS-F37 v1.3 and ZS-A31 v1.5.)